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===== MASTER TUTOR SCRIPT: MHF4U UNIT 3

- RATIONAL FUNCTIONS (TEACHING) Student: Rigden

Total Slides: 30

HOW TO USE THIS SCRIPT: Keep this text file open on a second monitor. The HTML presentation is completely clean and student-safe. Guide Rigden through the concepts using these prompts.

PART A: INTRODUCTION & THE BASICS

SLIDE 1: Title Slide [SAY] Welcome back, Rigden! Today we are officially starting Unit 3: Rational Functions, Equations, and Inequalities. [SAY] In Unit 1 and 2, we looked at polynomials—smooth, continuous curves. Today, we are looking at what happens when you divide one polynomial by another. The graphs are going to break, shatter, and shoot off into infinity. It's going to be a fun one!

SLIDE 2: Today's Roadmap [SAY] Here is our battle plan for today. First, the anatomy of a rational function. Second, the difference between a Vertical Asymptote and a Hole. Third, predicting Horizontal and Slant Asymptotes. And finally, solving equations and inequalities.

SLIDE 3: Warm-up (The Reciprocal Function) [SAY] Before we look at the complex Grade 12 stuff, let's look at the grandfather of all rational functions from Grade 11: $y = 1/x$. [ASK] If you look at this graph, what happens when $x = 0$? Why is the graph split in half? [EXPECT] You can't divide by zero, so there is a vertical asymptote. [SAY] Exactly. The golden rule of math: Thou shalt not divide by zero. That rule drives this entire unit.

SLIDE 4: What is a Rational Function? [SAY] A rational function is just a fraction where the numerator is a polynomial, and the denominator is a polynomial. $f(x) = P(x) / Q(x)$. [SAY] The most important part of this entire fraction is the bottom: $Q(x)$. Whatever makes $Q(x)$ equal zero is a major restriction on our domain.

PART B: VERTICAL ASYMPTOTES VS. HOLES

SLIDE 5: The Two Types of Breaks [SAY] When the denominator equals zero, the graph breaks. But it can break in one of two ways. [SAY] It can be a massive invisible wall called a Vertical Asymptote, where the graph shoots up or down forever. OR, it can just be a tiny missing pixel called a Point Discontinuity, or a "Hole".

SLIDE 6: How to tell the difference? [SAY] How do we know which one it is? We have to factor everything first. [SAY] Here is the rule: If a factor in the denominator CANCELS OUT with a factor in the top, it creates a Hole. If the factor is trapped and CANNOT cancel, it creates a Vertical Asymptote.

SLIDE 7: Worked Example (Hole vs VA) [SAY] Let's look at $f(x) = (x^2 - 4) / (x - 2)$. [ASK] What is the very first step we should take here? [EXPECT] Factor the top. Difference of squares: $(x-2)(x+2)$. [SAY] Spot on. Now look what happens. We have an $(x-2)$ on top and bottom. They cancel out! [ASK] Because they canceled, what is happening at $x = 2$? [EXPECT] It's a hole! [SAY] Yes! The graph will literally just look like the line $y = x + 2$, but with a tiny hole poked out of it at $x = 2$.

SLIDE 8: Finding the Hole's Coordinates [SAY] We know the hole is at $x = 2$. But what is its y -coordinate? [SAY] You can't plug 2 into the original equation because you'll get $0/0$. So, we plug it into the SIMPLIFIED equation: $y = x + 2$. [ASK] If we plug 2 into $y = x + 2$, what do we get? [EXPECT] $y = 4$. [SAY] Exactly. The hole is at the coordinate $(2, 4)$.

SLIDE 9: Your Turn! [SAY] Grab your paper. Try this one: $g(x) = (x^2 + 3x) / (x^2 - 9)$. [ASK] Factor the top, factor the bottom, and tell me: Where is the VA, and where is the Hole? [EXPECT] Top is $x(x+3)$. Bottom is $(x-3)(x+3)$. The $(x+3)$ cancels, so there is a hole at $x = -3$. The $(x-3)$ stays, so there is a VA at $x = 3$.

SLIDE 10: Your Turn (Solution) [SAY] Check your work against the board. Excellent job identifying the canceled factor versus the remaining factor.

PART C: HORIZONTAL & SLANT ASYMPTOTES

SLIDE 11: End Behaviour (Horizontal Asymptotes) [SAY] We know what happens when the denominator is zero. But what happens when x gets massively huge? What happens at positive and negative infinity? [SAY] This is called End Behaviour, and it gives us our Horizontal Asymptotes (HA). Instead of factoring, we only care about the HIGHEST DEGREE on the top and bottom. It's a tug-of-war.

SLIDE 12: Rule 1 - Bottom Heavy [SAY] Rule 1: If the degree of the bottom is BIGGER than the top, the bottom wins the tug-of-war. [SAY] Imagine dividing 5 slices of pizza by a billion people. Everyone basically gets zero pizza. [SAY] So, if it's bottom-heavy, the HA is ALWAYS $y = 0$.

SLIDE 13: Rule 2 - Equal Degrees [SAY] Rule 2: What if it's a tie? What if the degrees are exactly the same? [SAY] Look at $(6x^2) / (2x^2)$. The x^2 terms basically tie each other. So we just divide the leading coefficients! [ASK] What is 6 divided by 2? [EXPECT] 3. [SAY] Exactly. The HA is $y = 3$.

SLIDE 14: Rule 3 - Top Heavy (Slant Asymptotes) [SAY] Rule 3: What if the top is exactly ONE degree higher than the bottom? Like an x^2 over an x . [SAY] The top wins, so there is NO horizontal asymptote. Instead, it creates an angled line called a Slant or Oblique Asymptote. [ASK] How do we find the equation of a line when dividing polynomials? What did we learn in Unit 2? [EXPECT] Long division or synthetic division!

SLIDE 15: Finding the Slant Asymptote [SAY] Let's find the slant asymptote for $y = (x^2 + 5x + 4) / (x - 2)$. [ASK] We are dividing by $(x - 2)$. What number goes outside our synthetic division bracket? [EXPECT] Positive 2. [SAY] Perfect. We run the drop-multiply-add loop. We get a quotient of $x + 7$, with a remainder of 18. [EXAM TIP] For slant asymptotes, we IGNORE the remainder! The equation of the slant asymptote is just the quotient: $y = x + 7$.

SLIDE 16: Asymptote Rapid Fire [SAY] Let's test these three rules. I'm going to show you three functions. Tell me if the HA is zero, a ratio, or if it's a slant asymptote. [ASK] Function A: $f(x) = (3x) / (x^2 + 1)$. Top degree is 1, bottom is 2. [EXPECT] Bottom heavy, so $y = 0$. [ASK] Function B: $g(x) = (8x^2) / (4x^2 - x)$. Degrees are tied. [EXPECT] Ratio. $8 / 4 = 2$. So $y = 2$. [ASK] Function C: $h(x) = (x^3) / (x^2 + 1)$. Top is exactly one bigger. [EXPECT] Slant asymptote.

PART D: SOLVING RATIONAL EQUATIONS & INEQUALITIES

SLIDE 17: Solving Rational Equations [SAY] Now that we know what they look like, let's solve them. [SAY] Look at $x / (x-2) = 5 / (x-2)$. [SAY] Since the denominators are already exactly the same, the math is easy: we just set the numerators equal to each other! So $x = 5$. [SAY] Always, always, always check your answer against the restrictions. Does $x = 5$ cause the denominator to be zero? No. So it's a valid answer.

SLIDE 18: The Extraneous Root Trap [SAY] But watch out for this trap. Let's solve $x / (x-3) = 3 / (x-3)$. [ASK] If we set the numerators equal, what does x equal? [EXPECT] $x = 3$. [SAY] Right. But look at the original denominator. What happens if we plug 3 into $(x-3)$? [EXPECT] We get 0. We can't divide by zero! [SAY] Exactly! Because our answer breaks the universe, we throw it out. It's an "extraneous root". This equation has NO solution.

SLIDE 19: Finding Common Denominators [SAY] What if the denominators aren't the same? $3 / (x+2) = 4 / (x-1)$. [SAY] We find a common denominator, or in this simple case, we can cross-multiply! [SAY] 3 times $(x-1) = 4$ times $(x+2)$. Expand it, solve for x , and check your restrictions.

SLIDE 20: Rational Inequalities (The Danger Zone) [SAY] Equations are fine. Inequalities are where students lose all their marks on the unit test. [SAY] Look at this: $(x - 2) / (x + 3) > 1$. [ASK] Your brain probably wants to cross-multiply the $(x+3)$ to the other side. But why is that a fatal error in an inequality? What is the rule about multiplying inequalities by negative numbers? [EXPECT] If you multiply by a negative, you have to flip the sign. [SAY] Exactly! And because ' x ' is a variable, we have no idea if $(x+3)$ is a positive number or a negative number! So we don't know whether to flip the sign or not.

SLIDE 21: The Safe Method for Inequalities [SAY] Because we can't cross-multiply variables, we must use the Safe Method. [SAY] Step 1: Bring EVERYTHING to one side so it's > 0 or < 0 . We subtract the 1 over. [SAY] Step 2: Get a common denominator to merge it into one giant fraction. [SAY] Step 3: Find the critical values (roots and VAs) and use an interval testing chart.

SLIDE 22: Interval Testing [SAY] Let's say we simplified an inequality to $(x-4) / (x+2) > 0$. [SAY] Our critical values are $x=4$ (from the top) and $x=-2$ (from the bottom). [SAY] We put these on a number line, which creates 3 zones. We pick a random test number in each zone to see if the fraction turns out positive or negative! If it's positive, it satisfies our " > 0 " condition.

PART E: CONSOLIDATION & WRAP-UP

SLIDE 23: Your Turn (Inequalities) [SAY] Let's see you try the safe method. Solve $(x+1) / (x-3) < 0$. [ASK] It's already less than zero, so no need to move things over. What are your two critical values for your number line? [EXPECT] Top gives $x = -1$. Bottom gives $x = 3$. [SAY] Great. Set up your number line. Test $x = -2$, test $x = 0$, and test $x = 4$. Tell me which zone gives you a negative result.

SLIDE 24: Your Turn Solution [SAY] Check your work! The only zone that gave a negative result (less than zero) was the middle zone. So the answer is between -1 and 3 .

SLIDE 25: Summary of Asymptotes [SAY] Let's lock in everything we learned today. [SAY] VAs happen when denominator factors don't cancel. Holes happen when they DO cancel. [SAY] HAs are a tug-of-war. Bottom heavy = 0. Tied = Ratio. Top heavy by 1 = Slant.

SLIDE 26: Summary of Equations/Inequalities [SAY] For equations, cross multiply or get a common denominator, but ALWAYS check for extraneous roots. [SAY] For inequalities, NEVER cross multiply variables. Bring it to one side and use a number line.

SLIDE 27: Exit Ticket [SAY] 3 rapid fire questions to get you out of here! [ASK] 1. True or False: A slant asymptote happens when the top degree is 2 and the bottom is 1. (True) [ASK] 2. Does $(x-2)/(x-2)$ have a VA or a hole at $x=2$? (Hole, because it cancels). [ASK] 3. Can you cross-multiply variables in an inequality? (NO!)

SLIDE 28: Homework [SAY] Amazing job today, Rigden. This unit is conceptually tough, but you handled it perfectly. [SAY] Your homework is to practice these rules. Use the "Practice Mastery" deck we created for you. It has 40 questions to test these exact concepts. See you next class!